



MIAMI UNIVERSITY

DISCOVERY CENTER

Evaluation • Research • Professional Learning

Evaluation of *SIGHT: Safety Immersion and Gamified Hazard Training for Industry 5.0 Workers*

ANNUAL REPORT 2026

DISCOVERY CENTER

for

EVALUATION, RESEARCH, AND PROFESSIONAL LEARNING

MIAMI UNIVERSITY

OXFORD, OH

Please cite as follows:

Morio, K., & Li, Y. (2026). *Evaluation of SIGHT: Safety Immersion and Gamified Hazard Training for Industry 5.0 Workers: Annual report 2026*. Oxford, OH: Miami University, Discovery Center for Evaluation, Research, and Professional Learning.

Distributed by:

© Discovery Center for Evaluation, Research, and Professional Learning
Miami University, Oxford, OH

408 McGuffey Hall
210 E. Spring St.
Oxford, Ohio 45056
discoverycenter@MiamiOH.edu
(513) 529-1686 phone
(513) 529-2110 fax

Table of Contents

Table of Contents.....	3
Table of Tables	4
Executive Summary.....	Error! Bookmark not defined.
Evaluation Methods.....	Error! Bookmark not defined.
Key Evaluation Findings, Project Year 1.....	Error! Bookmark not defined.
Project Aim 1.....	Error! Bookmark not defined.
Project Aim 2.....	Error! Bookmark not defined.
Introduction	5
Evaluation Methods and Findings.....	7
SIGHT Key Personnel Interview	7
BmE Student Summer Program Observation Data Collection.	Error! Bookmark not defined.
BmE Summer Teacher Workshop Observation Data Collection.....	Error! Bookmark not defined.
BmE Teacher Focus Group Data Collection	Error! Bookmark not defined.
BmE Key Personnel Interviews Data Collection.....	Error! Bookmark not defined.
Summer Program Document Review	7
Qualitative Data Analysis	Error! Bookmark not defined.
Qualitative Findings	Error! Bookmark not defined.
Project Goal 1.....	Error! Bookmark not defined.
Project Goal 2.....	Error! Bookmark not defined.
Summary and Recommendations.....	23
Summary	23
Recommendations	24
Appendices.....	25

Table of Tables

Table 1. Project Goals, Evaluation Questions, and Evaluation Activities.....	Error! Bookmark not defined.
Table 2. <i>Personal Teacher Efficacy and Beliefs, BmE Teacher Post Summer Workshop Questionnaire, Summer 2025 (n=5)</i>	Error! Bookmark not defined.
Table 3. <i>Teaching Outcome Expectancy Beliefs, BmE Teacher Post Summer Workshop Questionnaire, Summer 2025 (n=5)</i>	Error! Bookmark not defined.
Table 4. <i>Student Technology Use, BmE Teacher Post Summer Workshop Questionnaire, Summer 2025 (n=5)</i>	Error! Bookmark not defined.
Table 5. <i>STEM Instruction, BmE Teacher Post Summer Workshop Questionnaire, Summer 2025, (n=5)</i>	Error! Bookmark not defined.
Table 6. <i>21st Century Learning Attitudes, BmE Teacher Post Summer Workshop Questionnaire, Summer 2025, (n=5)</i>	Error! Bookmark not defined.
Table 7. <i>Teacher Leadership Attitudes, BmE Teacher Post Summer Workshop Questionnaire, Summer 2025, (n=5)</i>	Error! Bookmark not defined.
Table 8. <i>STEM Career Awareness, BmE Teacher Questionnaire, Summer 2025, (n=5)</i>	Error! Bookmark not defined.
Table 9. <i>Time Spent in Summer Teacher Workshop, BmE Teacher Post Summer Workshop Questionnaire, Summer 2025, (n=5)</i>	Error! Bookmark not defined.
Table 10. <i>Summer Teacher Workshop Content and Usefulness, BmE Teacher Post Summer Workshop Questionnaire, Summer 2025, (n=5)</i>	Error! Bookmark not defined.

Introduction

The Discovery Center for Evaluation, Research, and Professional Learning (Discovery Center) completed the first year of a two-year evaluation of the *SIGHT: Safety Immersion and Gamified Hazard Training for Industry 5.0 Workers* project (the SIGHT project). The SIGHT project is funded by the Ohio Bureau of Workers' Compensation (BWC) and led by a team of researchers at Miami University (MU) in collaboration with industry partners. The project's long-term goal is to reduce preventable workplace injuries across Ohio's manufacturing sector by establishing an adaptive safety training ecosystem through two complementary aims:

1. Develop a dynamic, web-based virtual reality (VR) platform for gamified hazard recognition training.
2. Implement and validate augmented reality (AR)-based coaching modules to support safety and productivity in manufacturing.

The project consists of five phases designed to develop and validate a proof-of-concept ecosystem that integrates web-based VR training with AR-enabled real-time guidance and targets BWC priority areas related to manufacturing hazards:

- Phase 1: Data Collection and Organization for Enhancing the Quality of Generative AI Outputs
- Phase 2: Core AI Development
- Phase 3: Creating the Gamified Training Script and Analytics Layer
- Phase 4: Developing the XR Proof-of-Concept
- Phase 5: Testing and Fine-Tuning the Proof-of-Concept XR Ecosystem

The mixed-methods evaluation includes formative, summative, and transformative components designed to assess project implementation and outcomes and support ongoing improvement. The formative component provides timely feedback to help the project team monitor implementation progress and identify needed adjustments to project activities. The summative component assesses the extent to which the project achieves its intended aims and associated outcomes. The transformative component identifies successes and opportunities for improvement to help ensure that the ecosystem developed through the SIGHT project works effectively for users across all relevant differences. The following evaluation questions (EQs) guided the evaluation activities.

The Formative Evaluation Questions (FEQs) are:

- FEQ1: Did the project team complete tasks in the expected time frame for each phase?
- FEQ2: Did the project team meet success metrics outlined for each phase?
- FEQ3: What unanticipated challenges threatened success in each phase?

- FEQ4: What unexpected facilitators supported success in each phase?

Summative Evaluation Questions (SEQs) are:

- SEQ1: Did the dataset created to power the AI engine support the AR and VR components in the ways expected by the project team, industry partners, and pilot testers?
- SEQ2: Did the project result in a VR training platform that is accessible to anyone in Ohio?
- SEQ3: Did the project result in an AR-based system that provides real-time, hands-on guidance in hazard recognition, safe machine operation, and optimized work practices?

Transformative Evaluation Questions (TEQs) are:

- TEQ1: In Phases 2, 4, and 5, did expert review, user testing, and / or pilot testing reveal any meaningful patterns in errors or inaccuracies?
- TEQ2: If meaningful patterns in errors or inaccuracies were identified, in what ways and how well did the project team address these patterns?
- TEQ3: In Phase 3, did the project team engage stakeholder and reviewer groups with the appropriate diversity and expertise to provide the feedback needed for thorough script validation?
- TEQ4: In Phase 3, To what extent did stakeholders and reviewers believe their feedback was useful to and valued by the project team?

Evaluation Methods

External evaluation activities in Project Year 1 included:

- submitting the IRB application for evaluation data collection activities;
- participating in and observing all 21 biweekly project meetings held since the project's inception;
- conducting seven key personnel interviews with the lead faculty principal investigator (PI), two Co-PIs, two project consultants, and two students;
- reviewing project development and research materials stored in the shared project Google Drive;
- tracking project progress across the 15 proposed tasks and comparing progress against the nine success metrics specified in Special Condition #1, WSIC-3 Section 2.5.5 (p. 24 of Attachment D to the WSIC Contract);
- analyzing evaluation data; and
- maintaining regular communication between the Evaluation Team and Project Team.

Meeting Observations and Document Review

Between July 1, 2025, and June 30, 2026, the Evaluation Team participated in and observed all 21 project team meetings held during Project Year 1. In June 2026, the Evaluation Team also conducted a systematic review of documents related to project implementation, product development, research, and dissemination. Materials reviewed included meeting agendas, presentations, and minutes, as well as relevant project development and research materials. All materials were accessed through the shared project Google Drive.

Task Progress and Success Metrics

The Evaluation Team used a Task and Metric Checklist to systematically track progress toward completion of the project's proposed tasks and achievement of the associated success metrics. To assess progress, the Evaluation Team reviewed multiple sources of evidence, including biweekly project meeting observation notes, meeting presentations and minutes, and project deliverables and other relevant project materials. Evidence across these sources was triangulated to assess the extent to which tasks were completed within the expected timeframes and success metrics were achieved.

SIGHT Key Personnel Interview Data Collection and Analysis

In May and June 2026, Evaluation Team members conducted seven key personnel interviews with members of the Project Team. Interviewees included the project's lead PI, two Co-PIs, two project consultants, and two students. Collectively, the interviewees represented a range of

roles and perspectives across the Project Team, providing broad representation of the team as a whole.

The interview protocols (see Appendix A) were developed by the Evaluation Team to explore the experiences of project team members and gather qualitative data to inform the Formative, Summative, and Transformative Evaluation Questions. Six interview protocols were developed to reflect the diverse roles of project stakeholders, including Principal Investigators and Co-Principal Investigators, consultants, project management personnel, individuals involved in the virtual reality (VR) training system, individuals involved in the augmented reality (AR) training system, and student team members. While each protocol was tailored to the participant's role and responsibilities, all included questions related to project challenges and successes, review and feedback processes for key project components (e.g., Retrieval-Augmented Generation [RAG], VR, and AR), collaboration and communication among project team members, student professional development, and priorities for subsequent years of the project.

Interview invitations were distributed on a staggered schedule based on participant role. Invitations to the lead PI and Co-PIs (Co-PIs) ($n = 6$) were initially sent on April 3, 2026, with follow-up reminders issued on April 15, April 30, and May 6 to individuals who had not yet responded. Invitations to student team members ($n = 4$) were initially distributed on April 24, 2026, with reminder invitations sent on April 30 and May 6. Invitations to consultants and industry partners ($n = 5$)¹ were distributed on May 6, 2026.

All interviews were conducted via Zoom and ranged from 20 to 68 minutes in length. With participants' permission, interviews were audio recorded and professionally transcribed using Rev.com. The evaluation team reviewed each transcript for accuracy prior to analysis.

One Evaluation Team member uploaded interview and focus group transcripts into the data analysis software NVivo for further analysis. Seven deductive codes were applied to the data based on a codebook developed by the Evaluation Team. The deductive codes reflected the evaluation questions posed at the outset of the project. Code definitions were refined as the data were analyzed. In the second round of analysis, inductive codes were applied to data within each deductive code to further elucidate the nature of the data collected under each code. Table 1 presents the code names, acronyms, and operational definitions, while Table 2 summarizes the number of interviews in which each code was identified and the total number of references associated with each code. In the third round of analysis, inductive codes were combined to reflect emergent themes. These data were analyzed using Nvivo data analysis software (version 14). The findings are presented in the below section.

¹ The industry partner, MeetKai, was invited to participate as a single focus group; therefore, one invitation was sent on behalf of the group rather than to individual participants.

Table 1. *Codebook for the Key Personnel Interviews, Summer 2026*

Code (Acronym)	Definition
Challenges (CHA)	Factors internal and/or external to the project that have served as obstacles to successful completion of project goals, objectives, and/or metrics. Discussions about ways to address challenges.
Facilitators (FAC)	Factors internal and/or external to the project that have supported successful completion of project goals, objectives, and/or metrics.
Wins (WIN)	Elements, activities, and/or accomplishments of the project team members identify as wins for the project overall. Benefits of the project to the participant's development, educational, professional, or otherwise.
Patterns (PAT)	Patterns in system performance revealed in the development, testing, and/or validations phases of project systems.
Adjustments (ADJ)	Adjustments made to address errors or challenges revealed in the development, testing, and/or validation stages of project systems. Discussions about plans for improvements in future years and/or iterations of the project.
Stakeholder Feedback (FBK)	Descriptions of the process to obtain, the application of, and/or the usefulness of stakeholder (i.e., non-key personnel) feedback.
Value (VAL)	Discussions, descriptions, or examples of the expertise team members use to enhance project success.

Table 2. *Number of References for Interview Codes*

Code (Acronym)	# of Interviews	# of References
Challenges (CHA)	7	49
Facilitators (FAC)	5	28
Wins (WIN)	7	46
Patterns (PAT)	4	5
Adjustments (ADJ)	4	12
Stakeholder Feedback (FBK)	3	14
Value (VAL)	7	33

The Evaluation Team developed a crosswalk aligning the EQs with data sources and analysis methods/interview codes (Table 3). The following section presents the evaluation findings organized by these evaluation questions.

Table 3. *Evaluation Question, Data Sources, and Analysis/Codes*

EQ#	Evaluation Question	Data Source(s)	Analysis/Codes
FEQ1	Did the project team complete tasks in the expected time frame for each phase?	• Task and Metric Checklist • Meeting Materials • Project Materials	• Number/percent completed • Number/percent completed within anticipated timeframe
FEQ2	Did the project team meet success metrics outlined for each phase?	• Task and Metric Checklist • Meeting Materials • Project Materials	• Number/percent completed • Number/percent completed within anticipated timeframe
FEQ3	What unanticipated challenges threatened success in each phase?	• Key Personnel Interviews • Meeting Materials	Interview codes: • CHA
FEQ4	What unexpected facilitators supported success in each phase?	• Key Personnel Interviews • Meeting Materials	Interview codes: • FAC
SEQ1	Did the dataset created to power the AI engine support the AR and VR components in the ways expected by the project team, industry partners, and pilot testers?	• Key Personnel Interviews • Meeting Materials	Interview codes: • FAC • WIN
SEQ2	Did the project result in a VR training platform that is accessible to anyone in Ohio?	• Website review at the end of Year 2 • Meeting Materials • Project Materials	Access website to review platform
SEQ3	Did the project result in an AR-based system that provides real-time, hands-on guidance in hazard recognition, safe	• AM Hub site visit observation notes in Year 2	Access AM Hub to review systems

EQ#	Evaluation Question	Data Source(s)	Analysis/Codes
	machine operation, and optimized work practices?	• Meeting Materials • Project Materials	
TEQ1	In Phases 2, 4, and 5, did expert review, user testing, and / or pilot testing reveal any meaningful patterns in errors or inaccuracies?	• Key Personnel Interviews • Meeting Materials	Interview codes: • PAT
TEQ2	If meaningful patterns in errors or inaccuracies were identified, in what ways and how well did the project team address these patterns?	• Key Personnel Interviews • Meeting Materials	Interview codes: • ADJ
TEQ3	In Phase 3, did the project team engage stakeholder and reviewer groups with the appropriate diversity and expertise to provide the feedback needed for thorough script validation?	• Key Personnel Interviews • Meeting Materials	Interview codes: • FBK • VAL
TEQ4	In Phase 3, To what extent did stakeholders and reviewers believe their feedback was useful to and valued by the project team?	• Key Personnel Interviews	Interview codes: • FBK

Evaluation Findings

This section presents findings from the formative, summative, and transformative evaluations conducted during Project Year 1. Some evaluation questions were grouped under broader categories to reduce duplication in reporting.

Formative Evaluation Findings

FEQ1: Did the project team complete tasks in the expected time frame for each phase?

The Evaluation Team fully agrees with the task completion dates reported by the Project Team and presented in Section 4, “Interim Activity Status,” of the DO8.1 Interim Progress Report.

In Project Year 1, the SIGHT project completed Project Phases 1-3: 1) Data Collection and Organization for Enhancing the Quality of Generative AI Outputs; 2) Core AI Development; and 3) Creating the Gamified Training Script and Analytics Layer. The project team also made progress on Phases 4-5: 4) Developing the XR Proof-of-Concept and 5) Testing and Fine-Tuning the Proof-of-Concept XR Ecosystem.

Evaluation findings indicate that the project team completed all tasks within the expected timeframe for each phase, with several tasks completed ahead of schedule. These findings are based on multiple data sources, including the task completion checklist; observations of biweekly project meetings and reviews of associated presentations and meeting notes; interviews with key project personnel; and reviews of project materials maintained in the project’s shared Google Drive.

By September 2025, the team had completed Phase 1 Tasks 1.1-1.3 by compiling equipment manuals and relevant federal safety regulations and guidance, creating an image database that substantially exceeded the original target, and completing detailed metadata tagging. The team also made progress ahead of schedule on later-phase activities, including developing and deploying a functional RAG chatbot (Phase 2 Task 2.1), developing image classification tools (Phase 2 Task 2.2), and initiating 3D equipment rendering (Phase 4 Task 4.2).

By December 2025, the team had continued to complete planned activities on time while advancing several tasks beyond their anticipated stages of development. The RAG model was fully developed, rigorously benchmarked, and deployed through a production-ready API (Phase 2 Task 2.1), while the image classification model (Phase 2 Task 2.2) achieved 99.6% mean average precision. For Phase 3, the team developed early functional VR training prototypes while continuing work on training scripts (Task 3.1), initiated development of the AR interface and user experience (Task 3.2), and began establishing the analytics framework (Task 3.3). The

team also established the data acquisition and connectivity foundation for the digital twin environment (Phase 4 Task 4.1), advanced 3D rendering through two complementary development approaches (Task 4.2), and deployed an initial speech-to-speech capability while evaluating options for full integration (Task 4.3).

By March 2026, the team had completed and deployed all 18 gamified training modules across the three target machines (Phase 3 Task 3.1), implemented the integrated gamification engine (Task 3.2), and completed the analytics framework (Task 3.3). The team also completed high-fidelity 3D equipment renders through both planned development routes (Phase 4 Task 4.2) and completed real-time speech-to-speech AI integration (Task 4.3). In addition, the team continued to make substantial progress on the digital twin environment (Task 4.1), reaching approximately 60% completion and developing two-way digital twins for two of the three target machines.

By June 2026, the team had nearly completed the digital twin environment (Phase 4 Task 4.1), with only minor testing and refinement remaining. The integrated VR and AR training system (Task 4.4) had reached approximately 50% completion, and Phase 5 tasks were progressing on schedule.

Overall, the available evidence indicates that the project team has maintained strong alignment with the expected implementation timeframe. Tasks scheduled for the earlier phases were completed on time or ahead of schedule, and the team frequently initiated later-phase work early to reduce downstream rework and accelerate integration. The remaining work is concentrated in later-stage integration activities, particularly finalizing the digital twin environment and completing the fully integrated VR and AR training system, both of which were actively progressing by the end of Project Year 1.

FEQ2: Did the project team meet success metrics outlined for each phase?

Evaluation findings indicate that the project team has met the first six success metrics, including all five metrics for Project Phases 1-3 and one metric for Phase 4, either on time or ahead of schedule (Table 4). The project team is also making timely progress toward the remaining three success metrics across Phases 4 and 5. These findings are based on multiple sources of evaluation data, including observations of biweekly project meetings and reviews of associated presentations and meeting notes; interviews with key project personnel; and reviews of project materials maintained in the project's shared Google Drive.

The project not only met the timelines associated with these success metrics but also exceeded several originally proposed targets. For example, in Phase 1, the project substantially exceeded Success Metric 2, which called for creating an image database containing at least 60 high-quality, metadata-tagged images per machine. By September 2025, the team had created a dataset of approximately 6,000 equipment images, including approximately 1,200 original

images of the three target machines and additional synthetic images generated through transformations such as rotation, cropping, noise addition, and bounding-box rotation.

For Phase 2 Success Metric 3, the team developed, benchmarked, and deployed a RAG-based safety chatbot grounded in the operational and safety materials compiled during Phase 1. The team evaluated 24 RAG pipelines using an expert-validated benchmark covering the three target machines and selected a final pipeline that achieved 86.66% accuracy, exceeding the established threshold of 80% response accuracy for safety-critical information queries.

For Phase 2 Success Metric 4, the team developed and trained a YOLOv11-based machine recognition system using the image dataset created during Phase 1. The model achieved 99.6% mean average precision and demonstrated strong performance across machines, operators, and lighting conditions. This performance substantially exceeded the established thresholds of 80% classification accuracy under standard lighting and 75% accuracy under varying environmental conditions.

For Phase 3 Success Metric 5, the team completed and deployed a functional analytics framework for the web-based VR training platform, with data from training events captured in under one second following each event. This performance exceeded the success criterion of capturing data from at least 90% of training interactions within five seconds per interaction.

Overall, the findings demonstrate strong progress toward the phase-specific success metrics during Project Year 1 and indicate that project implementation is generally proceeding ahead of the anticipated timeline.

Table 4. *Success Metrics and Associated Status*

Success Metric and Description	Status	% of Work Completed	Projected Completion Date
1. Compile 100% standardized operational and safety documentation for the three selected target machines (Bridgeport Manual Mill, TL-1 CNC, Universal Cobot).	Completed	100%	9/30/2025
2. Create a comprehensive image database containing a minimum of 60 high-quality, metadata-tagged images per machine for the three target machines, covering multiple angles and operational scenarios.	Completed	100%	9/5/2025
3. Achieve at least 80% response accuracy for safety-critical information queries with the Retrieval-Augmented Generation (RAG) system, validated through expert review and user testing.	Completed	100%	12/18/2025

Success Metric and Description	Status	% of Work Completed	Projected Completion Date
4. The AI image classification system will achieve at least 80% classification accuracy on a holdout dataset under standard lighting and at least 75% accuracy under varying environmental conditions.	Completed	100%	9/12/2025
5. Implement a functional analytics framework for tracking user interactions that successfully captures data from at least 90% of training interactions within 5 seconds per interaction.	Completed	100%	3/31/2026
6. Have functional VR and AR proof-of-concept prototypes that successfully integrate AI systems (RAG, image classification, speech-to-speech), display 3D renders/AR overlays for target equipment, and connect to the analytics backend.	Completed	100%	6/30/2026
7. Ensure XR prototypes achieve initial usability, demonstrated by first-time users (research team/safety consultant) completing setup and initiating use of both VR and AR systems in under 30 minutes each.	On Schedule	50%	12/31/2026
8. Successfully recruit participants balanced across student trainees and industry professionals. Ensure that at least 36 participants complete and evaluate all required scenarios in both VR and AR testing sessions.	On Schedule	20%	6/30/2027
9. Deploy the refined web-based VR application for public access via the project website, with the link provided in the final report.	On Schedule	10%	6/30/2027

FEQ3: What unanticipated challenges threatened success in each phase?

Given the project's large interdisciplinary team, involvement of partners from multiple sectors, ambitious goals, tight timelines, and specific deliverable requirements, the project has made strong progress while also encountering several implementation challenges. Six primary themes emerged under the challenges (CHA) code from the key personnel interviews, highlighting both operational and organizational factors that influenced project progress. Although participants described several obstacles, they also emphasized the project team's ability to adapt, address emerging challenges, and continue making progress toward project objectives.

Theme 1: Ensuring the project stayed on pace

The most frequently discussed challenge was maintaining project momentum and meeting established timelines. All members of the project leadership team acknowledged that keeping

the project on schedule required continuous monitoring and adjustment through the first year. Participants noted that the pace of the project, coupled with strict deliverable deadlines, required significant coordination and flexibility to ensure milestones were achieved. For instance, one participant stated,

“The biggest challenges is seeing the true end of it... The progress that we’ve made with that in a couple of months alone really changed it. And like, okay, this is a lot more doable than I thought it was because it felt slow progress at first early on and I see it just accelerating.”

Theme 2: Variability in personnel and partner contributions

Project leadership consistently described variability in the amount and quality of time contributed by project personnel and external partners as a significant implementation challenge. Participants explained that differences in availability and engagement occasionally shifted additional responsibilities to already heavily committed team members. As a result, project staff often assumed extra work to ensure that deliverables were completed, and BWC deadlines were met. When specifically asked about challenges, one Co-PI indicates this workload variability by stating, “...to be completely honest, different degrees of contribution. Some members are going above and beyond our expectations, others not as much, so we have to fill in the gaps all over the place.”

Theme 3: Meeting BWC expectations

Two participants discussed the expectations associated with the BWC grant and its deliverables. They noted that the project’s scope, milestone-based structure, and deliverable requirements required substantial planning, coordination, and attention to quality throughout implementation.

Theme 4: University infrastructure and administrative support

Participants also identified challenges related to university infrastructure, particularly pre-award and post-award administrative support. Two participants expressed the view that existing institutional processes were not fully equipped to support faculty managing large-scale, externally funded projects. One participant noted that, although project personnel and others across the university worked hard to address organizational challenges and maintain high-quality project outcomes, additional institutional infrastructure and support were needed. These administrative limitations placed additional demands on project personnel, who often assumed responsibilities beyond their anticipated project roles.

Theme 5: Translating the product to a broader manufacturing audience

Two participants identified a potential challenge for the second year of the project: ensuring that the project’s products and resources are applicable across the diverse manufacturing sector.

Theme 6: Recommendations for future project years

In addition to describing challenges encountered during the first year, two participants offered recommendations for strengthening project implementation in subsequent years. Suggested improvements focused on refining project processes, enhancing coordination among personnel and partners, and proactively addressing challenges identified during the initial implementation period. Participants viewed these lessons learned as opportunities to improve efficiency and support continued project success.

FEQ4: What unexpected facilitators supported success in each phase?

Three primary themes emerged under the code “facilitators” (FAC) from the interviews, highlighting the importance of effective leadership, strong collaborative relationships, and the availability of resources that supported project activities. Collectively, these facilitators helped the project team navigate challenges and maintain progress toward project goals and deliverables.

Theme 1: Principal Investigator leadership

Participants identified the leadership of the lead PI as an important facilitator of project success. Several participants described the PI’s ability to organize the team’s workflow, coordinate activities across the interdisciplinary team, monitor progress toward project milestones and deliverables, and respond to emerging challenges as instrumental in keeping the project moving forward. Participants also noted the substantial time and attention the PI dedicated to project management, tracking, and reporting, which helped maintain accountability and alignment with project expectations. One participant stated, “I will say that I’ve been really impressed with [the PI] his just project management and keeping everyone going,” while another stated, “So he dedicated a lot of time tracking and reporting. So I think that’s a significant amount of leadership.” Collectively, these perspectives suggest that consistent leadership and active project oversight played an important role in supporting coordination, maintaining momentum, and facilitating progress during Project Year 1.

Theme 2: Partnerships and collaboration

The most frequently discussed facilitator was the strength of partnerships and collaboration among project personnel and external partners. Participants emphasized that open communication, shared commitment to project goals, and the willingness of team members and partner organizations to work collaboratively strengthened project implementation. Two quotes illustrate this theme:

“The approach MeetKai brought to the project and the relationship they built with the team facilitated project outcomes.”

“One person who has no relationship to Miami at all. The other person thinks, ‘I’m a Miami alum and I think what you’re doing is cool and I just want to see what you’re doing.’ But both have been very gracious with sharing our Q&As with their

operators, with their engineers, and they played a really important role in our success.”

Theme 3: Resources that supported project implementation

Two participants identified the availability of project resources as an important contributor to successful implementation. Participants described how access to appropriate resources, including personnel, expertise, tools, and space, helped the project team accomplish key goals and objectives. These resources enhanced the team’s capacity to respond to challenges and supported the timely completion of project activities.

Summative Evaluation Findings

Findings from the summative evaluation reflect themes that emerged by the end of Project Year 1 rather than across the full two-year project period. Accordingly, findings for some SEQs are not yet available because the project is still in progress and has not yet reached its final phase.

SEQ1: Did the dataset created to power the AI engine support the AR and VR components in the ways expected by the project team, industry partners, and pilot testers?

Although interview participants were not asked directly whether the dataset created to power the AI engine supported the augmented reality (AR) and virtual reality (VR) components as intended, in addition to facilitators described above, responses to questions coded as project successes ("wins", WIN) provided indirect evidence related to this evaluation question. Overall, participants described successful completion and continued advancement of key project deliverables, suggesting that the dataset effectively supported development of the project's AI engine and associated training components. Three themes emerged under the “WIN” code that reflect what participants consider success from Year 1.

Theme 1: Project deliverables demonstrated successful development

The strongest evidence related to this evaluation question came from participants' descriptions of Year 1 project deliverables. Across interviews, team members identified the project's major technical accomplishments, including the AI chatbot, research products generated through the project, and the gamified training platform as significant project successes. For example, one participant stated, “The tech. I mean, the technology, the VR, the virtual reality one, it's almost done. It's like our plans to get it done, completely done, done, ready for testing, testing, testing and feedback by January next year... so that’s what I’m most proud of.”

Theme 2: Professional growth reflected successful project engagement

Participants also described substantial personal and professional benefits associated with working on the project. Team members reported gaining new technical knowledge, expanding their professional skill sets, and participating in opportunities that would not otherwise have

been available. Student team members highlighted the mentorship provided by the PI and Co-PIs, describing relationships that extended beyond traditional supervisory roles and contributed to their professional development. For instance, one student stated, “It’s not even mentorship, they’re friends. I’m [on a] first name basis with them, which is I don’t think a lot of students can say that about their relationship with professors.” Co-PIs similarly emphasized the knowledge, skills, and professional connections students developed through their involvement in the project. Although these findings do not directly assess dataset performance, they reflect a high level of engagement with and confidence in the project's development activities.

Theme 3: Networking and partnerships supported continued project success

Participants also identified networking opportunities and collaborative partnerships as important project successes. These relationships were viewed as beneficial not only for achieving current project objectives but also for supporting future collaborations, research initiatives, and career opportunities for project personnel. Participants suggested that these partnerships strengthened the project's long-term sustainability and expanded opportunities for continued innovation beyond the initial project period. For example, one participant described a potential partnership opportunity, stating:

“And that relationship has been wonderful. We're currently in the process of maybe submitting a DOE Genesis proposal... Yamaha is going to be our lead industry partner there and we're taking some of the amazing digital twin work that Ibrahim has been working on pitching it for the DOE proposal...”

SEQ2: Did the project result in a VR training platform that is accessible to anyone in Ohio?

Findings for SEQ2 will be available upon completion of the project.

SEQ3: Did the project result in an AR-based system that provides real-time, hands-on guidance in hazard recognition, safe machine operation, and optimized work practices?

Findings for SEQ2 will be available upon completion of the project.

Transformative Evaluation Findings

Similar to the summative evaluation, findings from the transformative evaluation reflect only themes that emerged by the end of Project Year 1 rather than across the full two-year project period. Accordingly, findings for some TEQs are not yet fully available because the project is still in progress and has not yet reached its final phase.

TEQ1: In Phases 2, 4, and 5, did expert review, user testing, and/or pilot testing reveal any meaningful patterns in errors or inaccuracies?

Interview participants were asked to reflect on whether review and user testing revealed meaningful patterns in errors or inaccuracies during project implementation. Interview transcripts were analyzed to determine the way in which participants described patterns (PAT) in system performance. Participants indicated that feedback obtained through these activities identified several recurring issues across different project components. Rather than representing isolated concerns, participants indicated these patterns provided valuable insight into areas requiring refinement and revisions throughout the development process.

Participants described how reviewers identified opportunities to improve the accuracy of project materials. The feedback enabled the project team to prioritize revisions, correct identified errors and inaccuracies, and streamline project products. One participant describes this process, stating:

“We come up with an answer from the model, ask human experts to come up with an answer to, and then we use some statistical techniques to see if semantically they are the same. And we realized that there was a bunch of questions, which that was not the case, which means that AI was wrong... So that's what we are looking at now, asking other people to also provide answers to that same question to see if a problem is with AI, or with the kind of ground truth answer...”

TEQ2: If meaningful patterns in errors or inaccuracies were identified, in what ways and how well did the project team address these patterns?

Interview participants were asked to reflect on how the project team adjusted (ADJ) in response to errors or inaccuracies identified during development. Findings indicate that the project team employed an iterative process of collecting, reviewing, and responding to stakeholder feedback to improve project products throughout implementation.

Two participants described the process used by MeetKai to collect feedback related to errors, inaccuracies, and opportunities for improvement. Participants explained that feedback from reviewers was frequently gathered, reviewed, and incorporated into subsequent revisions of project components. This iterative approach allowed MeetKai and the development team to identify issues as they emerged and make targeted adjustments to improve the functionality and usability of project products.

One participant described this by stating,

“We asked some people from industry and we asked some people from Miami through our classes. That is a form of evaluation that we know that not everybody evaluating that is an expert, but it gave us some feedback on usability related things

that I think was helpful. It allowed MeetKai to do quick iterations on some of their design choices, so I think that was quite helpful.”

Another reference indicated that the progress of the AR part of the project will determine if and in what ways the final experimental protocol that compares AR and VR systems will change.

TEQ3: In Phase 3, did the project team engage stakeholder and reviewer groups with the appropriate diversity and expertise to provide the feedback needed for thorough script validation? and TEQ4: In Phase 3, To what extent did stakeholders and reviewers believe their feedback was useful to and valued by the project team?

Interview findings provided indirect evidence related to Transformative Evaluation Questions (TEQ) 3 and 4. Although participants were not asked directly whether the stakeholder and reviewer groups reflected the appropriate diversity and expertise needed to support thorough script validation (TEQ3), interview responses provided insight into the project's feedback (FBK) processes and participants' perceptions of the expertise represented on the project team.

Participants described multiple opportunities to provide feedback on project components throughout development, suggesting that feedback was intentionally solicited at various stages of the project. One participant described the process used to collect and communicate feedback, noting that project team members were encouraged to review project materials and provide input for ongoing refinement.

Interview data also suggest that participants perceived their feedback as both useful and valued by the project team (TEQ4). Two participants specifically indicated that they never felt their comments or recommendations were overlooked and that feedback was consistently acknowledged and addressed during the development process.

Although interview data did not directly assess the diversity of stakeholder and reviewer groups, participants frequently referenced the breadth of expertise (VAL) represented among project personnel and collaborators. Three participants described the specialized knowledge and skills contributed by project colleagues. One participant highlighted the diverse perspectives and capabilities of the student team, while two participants specifically recognized the expertise provided by the project's postdoctoral researcher. Additional comments acknowledged the valuable contributions of external partners and consultants, whose expertise supported project development and implementation.

Additional Interview Finding: Priorities for Year 2

Although not directly aligned with a specific evaluation question, participants were asked to reflect on their priorities for the second year of the project. Responses from four interview participants provided insight into the team's perspective for the next year and the areas they

viewed as most critical for continued project success. Table 5 summarizes participants' perspectives on key plans and priorities for Year 2.

Table 5. *Priorities for Year 2 by Interviewee Role*

Role	Year 2 Priorities
Industry Consultant	Bring in additional industry leaders to the advisory board to provide feedback from a wider audience of professionals in manufacturing
Safety Consultant	Be more vocal and active in project meetings, particularly as it relates to the experimental protocol for comparing performance of the AR and VR systems. Helping to design, run, and analyze data from the experiment that compares the two systems.
Student	Completing the decision matrix for the robotic arm, and whatever else is needed to enable the machines (CNC, milling machine, robotic arm, pallet truck) to interface with humans in the AR environment.
Co-PI	VR pilot testing. Conversations related to expectations and compensation for all team members.

Summary and Recommendations

This section presents summaries of evaluation findings of the SIGHT project for Year 1 and recommendations for Year 2.

Summary

- The SIGHT project demonstrated strong overall progress, meeting implementation timelines and achieving or exceeding Year 1 success metrics. The Project Team completed all planned tasks within expected timeframes, with several activities initiated or completed ahead of schedule.
- The project also met the first six phase-specific success metrics and substantially exceeded several technical performance targets, demonstrating strong progress toward project goals.
- Year 1 technical accomplishments established a strong foundation for continued system integration, testing, and validation. Major accomplishments included development and deployment of the RAG-based safety chatbot, machine recognition system, gamified training modules, analytics framework, high-fidelity 3D equipment renders, speech-to-speech AI capabilities, and substantial portions of the digital twin and integrated XR environment. Collectively, these accomplishments provide evidence of meaningful progress toward the project's envisioned AI-enabled AR and VR training ecosystem (Project Aims 1 and 2).
- Strong leadership, collaboration, partnerships, and access to resources were important facilitators of project success. Active PI leadership supported coordination, progress monitoring, accountability, and responsiveness to emerging challenges. Productive relationships among project personnel, external partners, consultants, and industry collaborators, together with access to relevant expertise, personnel, tools, and space, further strengthened implementation.
- The project's ambitious scope and interdisciplinary structure created coordination, workload, and administrative challenges. Maintaining project pace required continuous monitoring and adjustment, while variability in personnel and partner contributions occasionally shifted additional responsibilities to highly engaged team members. Participants also identified limitations in university administrative infrastructure and support that increased demands on project personnel.
- The Project Team effectively used review, testing, and stakeholder feedback to identify and address errors and opportunities for improvement. Expert review and user feedback revealed recurring inaccuracies and areas requiring refinement across project components. Findings indicate that the team responded through an iterative process of reviewing feedback, investigating potential causes, and making targeted revisions to improve project products.

- Stakeholder feedback was viewed as useful and valued, although opportunities remain to broaden representation and industry input. Participants reported that feedback was consistently acknowledged and addressed and highlighted the substantial expertise represented among project personnel, students, consultants, and external partners.
- Project participation generated broader benefits related to professional development, research collaboration, mentorship, and networking. Participants described gains in technical knowledge and skills, access to new research opportunities, supportive mentoring relationships, and expanded professional connections.
- Year 2 priorities center on completing system integration, expanding testing and industry engagement, and strengthening team coordination. Participants identified priorities including VR pilot testing, continued AR and machine integration, broader industry advisory participation, increased involvement in experimental planning, and clearer expectations and compensation arrangements for project personnel. These priorities align with the project's transition from component development toward integrated testing, validation, and refinement.

Recommendations

Based on evaluation findings from available data sources, the Evaluation Team offers the following recommendations for Project Year 2.

- Prioritize completion, integration, and rigorous testing of the project's core technical components. The Project Team should maintain its strong implementation momentum while focusing Year 2 efforts on completing the digital twin environment, integrating the VR and AR systems, conducting pilot testing, and allowing sufficient time for iterative refinement before final validation.
- Develop a coordinated Year 2 implementation and testing plan. The Project Team should translate identified Year 2 priorities into a clear plan that specifies responsibilities, timelines, dependencies, decision points, and mechanisms for monitoring progress, with particular attention to system integration, pilot testing, and the experimental comparison of AR and VR systems.
- Clarify team roles, expectations, workload distribution, and compensation arrangements. Project leadership should revisit expectations for personnel and partner contributions, regularly assess workload distribution, and proactively address participation gaps to reduce overreliance on highly engaged team members and support timely completion of project activities.

Appendices

Appendix A. <i>SIGHT Key Personnel Interview Protocol</i>	26
---	----